Input power connnector

J1
Input 12V DC

PWR_FLAG

VIN1

PGND

Battery connector

J3
Battery

BATTERY

PGND

Output power connnector

J2
Output 12V DC

SYS

PGND

IC1
BQ25798RQMR

STAT_1
VBUS_1
VBUS_2
BTST1
REGN
D+
D−
VAC2
VAC1_1
ACDRV2
ACDRV1
QON
CE
SCL
SDA
TS_1
ILIM_HIZ

BATP
BTST2
PROG
INT
BAT_1
BAT_2
SDRV_1
SYS
SW2
GND
SW1
PMID
STAT_2
VAC1_2
TS_2
SDRV_2

Q1
AON7528

Q2
AON7528

D1
12V

R1
270R

R4
100R

C23
47n 25V

R5
27k 1−2%

NT4

TP3
PGND

TP4
SYS

TP1
GND

boost side

BTST2
SW2
47n 25V
C1

L1
2.2uH

buck side

SW1
47n 25V
C2
BTST1

-INT
CHG_REQ
I2C_SCL
I2C_SDA
J4
Conn_01x06_Pin
MAIN_3V3

TS
ILIM_HIZ

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

REGN
R8
2k2
LED
D2
STAT
C3
4.7u 10V
NT2
ILIM_HIZ
R6
5k6
R7
33k
TS
NTC temp sensor
J6

Grounds

NT1
PWR_FLAG
GND
PGND
PWR_FLAG

PGND = power ground (thicker traces, special net on PCB)
GND = digital ground (ordinary traces)

PWR_FLAGs are for DRC in PCB editor.

C10 10u 25V
C4 10u 25V
C21 10u 25V
C19 10u 25V
C17 10u 25V
C14 10u 25V
C11 10u 25V
C5 100n 25V
C24 47u low ESR
C25 47u low ESR
C22 47u POSCAP
C20 47u POSCAP
C18 10u 25V
C15 10u 25V
C12 10u 25V
C6 100n 25V
C16 10u 25V
C13 10u 25V
C7 100n 25V
C8
1nF, 50V
C9
100n 25V

BAS116H
D3
REGN
CE_BIAS

BAS116H
D4
MAIN_3V3
CE_BIAS

R9
10k
CE_N
CHG_REQ
1R1
CHG_GATE
Q3
PMV16UN
R10
10k

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

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File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

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